PHY 275 — Classical Mechanics

Fall 2026 · MWF 2:00–2:50 PM · August 26 – December 17 · Stuart Hall 104

Instructor

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Student hours: Mon/Tue/Fri, morning until noon, or by appointment: https://calendar.app.google/22xRLLnu5xH8MQQw8

Department Chair: Firdevs Duru (FDuru@coe.edu)

Course Description

Studies Newtonian, Lagrangian, and Hamiltonian formulations of the laws of motion. Applications to systems of particles, extended objects, and oscillatory systems. Prerequisite: General Physics II & Laboratory (PHY-195/-195L).

Mathematical preparation. This course leans heavily on multivariable calculus, linear algebra, and ordinary differential equations. You are not expected to arrive fluent in all of it. Pre-class prep assignments in the first four weeks will refresh vector calculus, series expansion, complex exponentials, and first- and second-order ODEs. If any of that feels shaky, come see me in the first two weeks rather than the eighth.

Learning Objectives

By the end of this course, you will be able to:

1. Formulate and solve Newtonian equations of motion for particles and systems of particles, including motion with drag and motion observed from non-inertial frames. (Assessed: homework, in-class quizzes, midterm exam)
2. Derive the Euler–Lagrange equation from a variational principle, and explain the conceptual relationship between the calculus of variations and Lagrangian mechanics. (Assessed: in-class quizzes, final exam)
3. Construct Lagrangians and Hamiltonians for constrained systems using generalized coordinates, and obtain and interpret the resulting equations of motion. (Assessed: homework, final exam)
4. Implement and validate numerical solutions to mechanics problems in Python/Colab, including verification against analytic limits, conserved quantities, and known special cases. (Assessed: pre-class prep, homework, final project)
5. Communicate a modeling result in writing and orally, and document and critically evaluate any AI-assisted work you submit. (Assessed: final project proposal, tech demo, and presentation)

Required Texts and Course Materials

Classical Mechanics, J. R. Taylor, University Science Books. ISBN 1-891389-22-X.

Google Colab worksheets and other digital tools will be provided through the course Moodle page. Lecture slides and code are maintained on Moodle and GitHub.

How Grading Works in This Course

This course mirrors the professional engineering world. Your goal is not to be perfect on the first try — it is to iterate until the work is right. Every piece of work receives one of three marks:

- 0 (Retry) — the work is incomplete or does not yet meet the stated standard. You revise and resubmit. A retry is not a penalty; it is the normal path.
- 1 (Success) — the work meets the quality standard stated in the assignment. Every assignment lists what "meets the standard" means for that assignment.
- Exemplary — the work meets the standard and exceeds it, for example through exceptional documentation, an unusually clear argument, or a novel approach. An Exemplary mark is worth 1 point and also carries the Exemplary flag, which is used for plus grades.

Note on the standard. Points are awarded for work that meets the stated quality standard, not for effort alone. Effort is what gets you there, and unlimited retries mean sustained effort will get you there — but a submission is marked 1 when it is correct and complete, not when it is merely attempted.

Categories of work

- Homework (10 assignments, 1 point each). Typically, due one week after assignment. May be analytic problems, a numerical experiment, or a short report on a tool you tried out.
- Pre-class prep (8 assignments, 1 point each). Read an assigned section, refresh a math skill, or practice a numerical technique before we use it in class. Short by design.
- In-class quizzes (10 quizzes, 1 point each). Pen and paper, no devices, announced at least one class period in advance. These reflect the work you have already done on homework and prep. Each correct answer also earns a token (see Tokens, below).
- Midterm exam, Week 8 (11 questions, 1 point each). Covers Newtonian, dynamics, and Lagrangian.
- Final exam, Week 17 (11 questions, 1 point each). Cumulative, weighted toward Chapters 6–13.
- Final project (3 points: proposal, tech demo, presentation — 1 point each).
- In-class board work (no points, but the main source of tokens). Working a problem at the board in front of the class is not graded and never counts against you.

Homework and pre-class prep together form the coursework pool (18 points available). Quizzes and exams together form the quiz and exam pool (18 points available). Both pools contain more points than any grade requires, so a small number of misses will not sink your grade.

Revision and resubmission

- Homework and pre-class prep may be revised and resubmitted as many times as you need.
- Each resubmission must be submitted within one week of the date I return feedback, and must include a short paragraph at the top describing what you changed and why. A resubmission without that paragraph will be returned unread.
- Do not save your revisions for December. Work returned to you after December 4 may not leave enough time for a full revision cycle, and I cannot guarantee turnaround on anything submitted in the final week.

Tokens

Alongside points, work done in the room earns tokens.

Earning tokens

- 1 token for each correct answer on an in-class quiz.
- 1 token for taking a problem to the whiteboard and working it through for the class. Also counts: presenting a numerical result, walking us through a piece of code, or leading a worked example.
- Board work pays for a genuine attempt, not for being right. A wrong approach reasoned through out loud is often the most useful thing that happens in a class period, and it earns the token. There is no penalty of any kind for getting it wrong at the board.
- Maximum of one board-work token per person per class period, so that everyone gets turns.
- Homework, pre-class prep, and exams do not yield tokens. They earn points in the usual way.

Spending tokens

- 10 tokens: one homework or pre-class prep assignment is recorded as Success without you submitting it. Maximum of **three** redemptions of this type per semester.
- 5 tokens: a one-week extension on one homework. No limit beyond your balance.
- 1 token: an in-person skill-check to earn points if you missed a quiz or haven't earn the point for that subject matter.

Rules of spending tokens

- Redeem for a point before the assignment is due; to buy an extension, redeem before the original deadline.
- All redemptions must be made by the last day of week 17, Friday, December 11 9:00 pm. I will not process redemptions during finals week.
- All tokens and points will be tracked in spread-sheet and updated to students upon request.

Late work and extensions

- Extensions are handled through the token system above: use 5 tokens to get one extra week on a homework or project assignment.
- Beyond your tokens, late work is still accepted and can still earn 1 point, but it is not eligible for an Exemplary mark and I cannot promise timely feedback.
- The hard deadline for all work, including every resubmission, is Friday, December 18 at 9:00 PM. Nothing submitted after that time earns points.
- If something serious is going on in your life, talk to me. Extensions beyond the token system exist and are not difficult to arrange, but they have to be arranged.

Determining your final grade

Each row below gives the minimum number of points needed in each column for that grade. You must meet all three thresholds in the row.

Grade	Coursework points (18 available)	Quiz & exam points (18 available)	Final project points (3 available)
A	12	12	3
B	10	10	2

Grade	Coursework points (18 available)	Quiz & exam points (18 available)	Final project points (3 available)
C	8	8	1
D	5	5	0

- F is assigned if the requirements for a D are not all met.
- Plus grade. Add a plus to your letter grade if you meet all three thresholds for that grade and either (a) more than half of your recorded marks are Exemplary, or (b) you meet at least two of the three thresholds for the next grade up. A plus cannot be added to an A.
- Minus grade. If you fall short of exactly one threshold by exactly one point, you receive the higher letter grade with a minus. Falling short of one threshold by two or more points, or falling short of two or more thresholds, places you at the next grade down (where the plus and minus rules apply again).
- Examples. 12 / 22 / 3 → A− (one threshold short by one). 12 / 21 / 3 → B+ (short by two, so B, but two of the three A thresholds are met). 11 / 19 / 3 → B (two A thresholds missed, and only one A threshold met, so no plus).

Final Project

The final project asks you to model a mechanical system of your own choosing that is beyond what we solve analytically in class — a double pendulum, a spinning top, an orbit with perturbation, a driven nonlinear oscillator, a controlled inverted pendulum, or something you propose. You will build a numerical/physical model, verify it, and explain what it tells you.

- Proposal (due [Week 15]) — one page: the system, the coordinates you will use, the Lagrangian or equations of motion, and how you will check that your model is right.
- Tech demo (due [Week 16]) — working code or set-ups that integrates your equations of motion and at least one verification: a conserved quantity holding to a stated tolerance, a small-angle limit matching the analytic result, or a comparison to a published result.
- Presentation (Week 16) — 10 minutes plus questions, present in slide form.

Projects may be done individually or with team efforts (maximum group size: 3 students). Teamwork must submit a report break-down team effort and do a team-work peer review (The link will be provided by me). Each deliverable is marked 0 / 1 / Exemplary on the same basis as everything else, and proposal and tech demo may be revised once.

Attendance and Participation

Active participation in class is the most efficient way to learn this material; it saves you hours of self-teaching later.

Attendance is tracked but not graded. I will follow up with students who accumulate excessive absences, but no direct penalty is applied and you do not need to ask permission to miss a class.

Submitting Work

PDF is the preferred format, produced from Microsoft Word, Google Docs, or LaTeX. Code should be submitted as a Colab notebook link or an .ipynb file.

Handwritten derivations are entirely acceptable. Scan or photograph them clearly, in focus and right side up, and combine them into a single PDF. Illegible submissions will be returned as a retry.

Use of AI and Generative Tools

If you are unsure whether a particular use is acceptable, ask me before you submit.

General rule of thumb: AI tools are roller skates, not crutches. AI tools (ChatGPT, Gemini, Claude, Copilot, and similar) may be used as long as they support your learning and do not replace your independent thought or effort.

Allowed: using AI to explain a derivation you are stuck on, generate boilerplate or plotting code, suggest debugging strategies, or brainstorm project ideas.

Prohibited: submitting AI-generated content that you cannot explain, derive, or defend; submitting AI output as if it were your own work; using AI during in-class quizzes or exams.

Required documentation. Any assignment where you used AI must include an AI Appendix that shows you vetted and validated the output:

- Code — wrap AI-generated blocks in comments:
  ### Begin AI Code ###
  …
  ### End AI Code ###
- Images — caption clearly, e.g. [Image generated by (tool name)].
- Prompts — include the prompts that materially shaped your submission, as a log at the end.
- Validation — one or two sentences on how you checked the output was correct (dimensional analysis, a limiting case, a hand calculation, a numerical cross-check).

Tentative Schedule

This schedule is a plan, not a contract. I will adjust pace based on how the class is doing and will announce any changes in class and on Moodle.

Week	Topics	Taylor
1 (Aug 26–28)	Newtonian Foundations	Ch. 1
2 (Aug 31–Sep 4)	Newtonian Mech: First Order ODEs	Ch. 1–2
3 (Sep 7–11)	Eigenvalue and Eigenvectors, Higher Order Linear Systems	Ch. 10, 11
4 (Sep 14–18)	Intro to Vector Space	Ch. 11
5 (Sep 21–25)	Conservation Forces and Intro to Lagrange	Ch. 7
6 (Sep 28–Oct 2)	Lagrange Mech and Variational Calculus	Ch. 3, 6
7 (Oct 5–9)	System modeling with Lagrange mech	Ch. 4, 5, 7, 11
8 (Oct 12–16)	Review and Midterm Exam	-

Week	Topics	Taylor
9 (Oct 19–23)	Conservation Laws and Symmetry	Ch. 3, 4 7
10 (Oct 26–30)	Continue on Conservation Laws and Intro to Conjugated Momentum	Ch. 7
11 (Nov 2–6)	Virtual Work and Lagrange Multiplier part 1	Ch. 7, 10
12 (Nov 9–13)	Virtual Work and Lagrange Multiplier part 2	Ch. 8
13 (Nov 16–20)	Hamiltonian Mechanics and Phase Space	Ch. 12, 13
14 (Nov 23–27)	Thanks Giving Break	Ch. 10
15 (Nov 30–Dec 4)	Intro to Computational Method for System Modeling	-
16 (Dec 7–11)	Final Project Presentations	-
17 (Dec 14–17)	Final Exam	-

Please verify all dates against the official Coe College academic calendar; the calendar takes precedence over this table.

College-Wide Policies

Mission Statement

Coe College is a national, residential liberal arts college offering a broad array of programs in the arts, sciences, and professions. Our mission is to prepare students for meaningful lives and fulfilling careers in a diverse, interconnected world. Coe's success will be judged by the success of our graduates.

Academic Integrity

At Coe College, we expect academic integrity of all members of our community. Academic integrity assumes honesty about the nature of one's work in all situations. Such honesty is at the heart of the educational enterprise and is a precondition for intellectual growth. Academic dishonesty is the willful attempt to misrepresent one's work, cheat, plagiarize, or impede other students' academic progress. Academic dishonesty interferes with the mission of the College and will be treated with the utmost seriousness as a violation of community standards. Please refer to the Coe College Academic Catalog for complete information regarding Academic Integrity: www.coe.edu/academics/academic-resources/provosts-office/academic-integrity-policy

The course-specific policy on generative and assistive technology appears under "Use of AI and Generative Tools" above and governs this course.

Respectful Communication

Coe College's effective teaching and learning environment is built on respectful communication. This includes avoiding harmful language or behaviors and approaching any mistakes as an opportunity to learn how to do better together.

FERPA

Students should be aware of their rights regarding the privacy of their educational records. Detailed information about your rights can be found under the FERPA (Family Educational Rights and Privacy Act of 1974) section in the Academic Catalog and online here: https://www.coe.edu/academics/academic-resources/registrar/ferpa

In line with FERPA restrictions, students should be aware that an instructor cannot publicly post grades by student name, institutional student identification number, or social security number without first having obtained students' written permission.

The Definition of a Course Credit and Expected Workload

One course credit at Coe College constitutes 180 hours' worth of student work over the course of the term. "The Department of Education has defined one hour to be 50 minutes, so 150 60-minute hours is equivalent to 180 50-minute hours." This figure includes both the time spent in class and out of class completing course work. In other words, students are expected to devote a considerable amount of time outside of class to this course. For courses that meet in a standard MWF or T-Th slot, students should be expected to work seven hours a week outside of the three hours in class.

Requesting Academic Accommodations

Coe College is committed to creating a learning environment that meets the needs of its diverse student body. If you anticipate or experience any barriers to learning in this course, please visit my office hours to discuss these concerns with me.

If you have a medical, physical, psychological, or learning need documented by a healthcare provider, or experience an unforeseen obstacle during the semester, you can make an appointment with the Accessibility Services Coordinator to request an official accommodation under the Americans with Disabilities Act. You can find more information about the Accessibility Services Office by visiting the Learning Commons website or by emailing accessibility@coe.edu. If you have already been approved for accommodations through Accessibility Services, please meet with me so we can develop a plan for your success together.

Reporting of Sexual Assault or Misconduct

As an instructor, one of my responsibilities is to help create a safe learning environment on our campus. I also have a mandatory reporting responsibility related to my role as a faculty member. It is my goal that you feel able to share information related to your life experiences in classroom discussions, in your written work, and in any one-on-one meetings. I will keep the information you share with me private to the greatest extent possible. However, I am required to share information regarding sexual misconduct or students who may be in danger to themselves or to others. Students may speak to someone confidentially by contacting Student Life at 319-399-8843 or emailing titleix@coe.edu, Safety and Security at 319-399-8888, Emily Barnard (college counselor) at 319-399-8843, or visit Coe's Title IX website for more information.